

affirmoTM tutorial 5: results

The results object, the comparison, the CE plane and the output files

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Table of contents

1 The results object	1
2 Each arm	2
3 The comparison	3
4 The cost-effectiveness plane	4
5 The output files	4

This tutorial starts from a fresh analysis folder with the example input files, and runs the model (see tutorial 1):

```
library(affirmoTM)
```

```
tm_example_inputs()
results <- tm_run()
```

1 The results object

`tm_run()` returns the results and prints a summary:

```
results
```

<tm_results> mean per arm over 100 runs (discounted costs and QALYs)

	arm	patients	cost_total	qaly_total	intervention_cost_total
	<char>	<num>	<num>	<num>	<num>
1:	iabc	5000	65493591	18004.50	187494.6
2:	usual_care	5000	66035435	17980.58	0.0
	cost_per_patient	qaly_per_patient	intervention_cost_per_patient	ischaemic_stroke	
	<num>	<num>	<num>	<num>	<num>
1:	13098.72	3.601	37.499	429.23	
2:	13207.09	3.596	0.000	461.93	
	haemorrhagic_stroke	major_bleed	deaths		
	<num>	<num>	<num>		
1:	121.54	147.47	1850.85		
2:	129.50	160.45	1853.00		

iABC minus usual care (mean over runs, per patient):

Incremental cost: €-108.37 (intervention cost: €37.5)

```
Incremental QALYs: 0.00478
ICER: iABC dominant (cheaper, more QALYs)
Difference in events (whole cohort): ischaemic_stroke -32.7, haemorrhagic_stroke -8.0, major_bleed
↳ -13.0, deaths -2.1
```

Written to outputs/: summary.csv, summary_by_run.csv, comparison.csv, comparison_by_run.csv, ce_plane.png

The results are a list:

```
names(results)
```

```
[1] "by_run"          "summary"          "comparison_by_run" "comparison"
[5] "n_runs"          "currency"         "wtp"
```

Element	What it holds
by_run	One row per run and arm: the summary of each run
summary	The mean over the runs, one row per arm
comparison_by_run	iABC minus usual care, one row per run
comparison	The mean differences, the ICER and its quadrant
n_runs, currency, wtp	The settings used

2 Each arm

```
results$summary
```

```
      arm patients cost_total qaly_total intervention_cost_total
      <char>   <num>   <num>   <num>             <num>
1:    iabc    5000  65493591  18004.50             187494.6
2: usual_care  5000  66035435  17980.58              0.0
  cost_per_patient qaly_per_patient intervention_cost_per_patient ischaemic_stroke
      <num>           <num>             <num>             <num>
1:    13098.72         3.600901         37.49893         429.23
2:    13207.09         3.596116         0.00000         461.93
  haemorrhagic_stroke major_bleed deaths
      <num>           <num>   <num>
1:    121.54         147.47 1850.85
2:    129.50         160.45 1853.00
```

Column	Meaning
patients	Patients who enter within the horizon
cost_total, qaly_total	Discounted totals for the whole cohort
intervention_cost_total	The discounted intervention cost within cost_total
cost_per_patient	cost_total divided by the number of patients
qaly_per_patient	qaly_total divided by the number of patients
intervention_cost_per_patient	intervention_cost_total divided by the number of patients
one column per event	The years spent in that event's states (see events in settings.yaml)
deaths	The number of patients who died

Costs and QALYs are per patient; events and deaths are counts for the whole cohort.

3 The comparison

iABC minus usual care, as the mean over the runs:

```
results$comparison
```

```

delta_cost delta_intervention_cost delta_qaly      icer quadrant
  <num>          <num>          <num>      <num>   <char>
1:  -108.3687          37.49893 0.004784424 -22650.32 dominant
delta_ischaemic_stroke delta_haemorrhagic_stroke delta_major_bleed delta_deaths
          <num>          <num>          <num>          <num>
1:          -32.7          -7.96          -12.98          -2.15

```

- `delta_cost` and `delta_qaly`: the incremental cost and QALYs per patient.
- `delta_intervention_cost`: the part of `delta_cost` that is intervention cost.
- the other `delta_` columns: the difference in each event count and in deaths, for the whole cohort.
- `icer`: the mean incremental cost divided by the mean incremental QALYs.
- `quadrant`: where the result lies on the CE plane. It says how to read the ICER:

quadrant	iABC compared with usual care	Printed as
costlier, more QALYs	Costs more, gives more QALYs	€ ... per QALY gained
dominant	Costs less (or the same), gives more QALYs	iABC dominant
dominated	Costs more (or the same), gives fewer QALYs	iABC dominated
cheaper, fewer QALYs	Costs less, gives fewer QALYs	€ ... saved per QALY lost
no QALY difference	The same QALYs	undefined

The same differences for each run:

```
head(results$comparison_by_run)
```

```

      run seed delta_cost delta_intervention_cost delta_qaly delta_ischaemic_stroke
    <int> <num>      <num>          <num>          <num>          <int>
1:     1  2026   -96.34138          37.44662 0.004423595          -33
2:     2  2027  -108.60145          37.72581 0.002737998          -24
3:     3  2028   -85.63400          37.08563 0.002815770          -33
4:     4  2029   -82.62992          37.22436 0.002988407          -36
5:     5  2030   -91.03477          37.57388 0.007176810          -45
6:     6  2031  -109.03745          37.52646 0.008693408          -37
delta_haemorrhagic_stroke delta_major_bleed delta_deaths
          <int>          <int>          <int>
1:          -10           -1           -1
2:          -12           -11           0
3:           -2           -16          -1
4:           -3            -8          -3
5:            4           -11          -5
6:           -8           -20          -9

```

```
results$comparison_by_run[, .(mean = mean(delta_qaly), lowest = min(delta_qaly), highest =
  ↪ max(delta_qaly))]
```

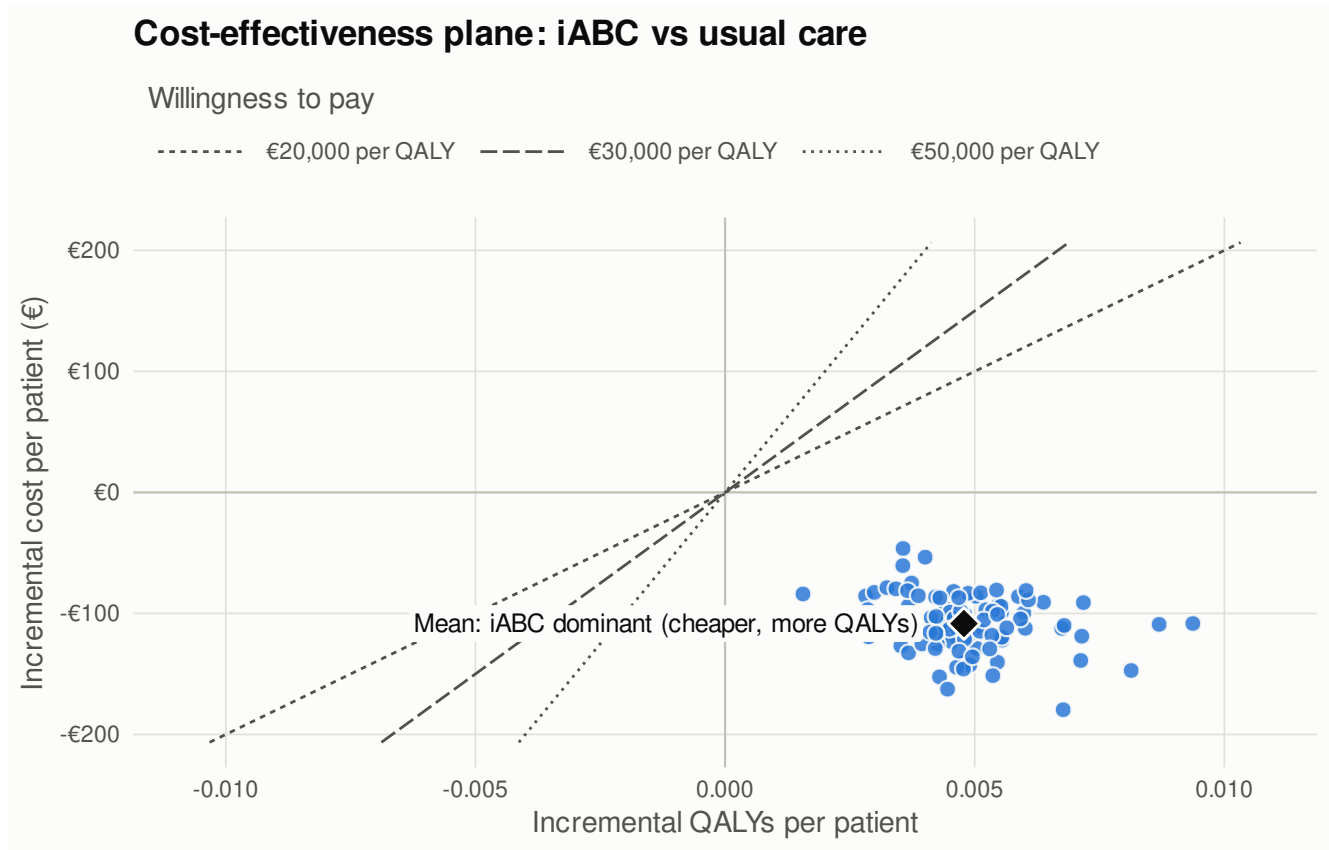
```

      mean      lowest      highest
    <num>    <num>    <num>
1: 0.004784424 0.00156039 0.009369034

```

4 The cost-effectiveness plane

```
plot(results)
```



- Each blue point is one run: incremental QALYs per patient across, incremental cost per patient up.
- The diamond is the mean across the runs, labelled with the result in the words of its quadrant.
- The dashed lines are the willingness-to-pay thresholds in **wtp**. Points below a line are cost-effective at that threshold.
- The runs use the same inputs and differ only in their random numbers, so the spread of the points shows how much a single run varies by chance.

`plot(results)` returns the plot, so it can be saved or changed with `ggplot2`:

```
p <- plot(results)
ggplot2::ggsave("my_ce_plane.pdf", p, width = 7, height = 5)
```

5 The output files

Every call to `tm_run()` writes these files to `outputs/`:

File	What it holds
<code>summary_by_run.csv</code>	One row per run and arm (<code>results\$by_run</code>)
<code>summary.csv</code>	The mean per arm (<code>results\$summary</code>)

File	What it holds
comparison_by_run.csv	iABC minus usual care, one row per run (results\$comparison_by_run)
comparison.csv	The mean differences, the ICER and its quadrant (results\$comparison)
ce_plane.png	The CE plane

```
list.files("outputs")
```

```
[1] "ce_plane.png"          "comparison_by_run.csv" "comparison.csv"
[4] "summary_by_run.csv"    "summary.csv"
```

```
fread("outputs/comparison.csv")
```

```

delta_cost delta_intervention_cost delta_qaly      icer quadrant
  <num>          <num>          <num>      <num>  <char>
1:  -108.3687          37.49893 0.004784424 -22650.32 dominant
delta_ischaemic_stroke delta_haemorrhagic_stroke delta_major_bleed delta_deaths
      <num>          <num>          <num>      <num>
1:          -32.7          -7.96          -12.98          -2.15
```